An Explainable CatBoost Framework for Loan Approval Prediction and Decision Support

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Abstract

Loan approval assessment involves the analysis of applicant information to estimate whether a loan application satisfies a lender's eligibility criteria. Machine learning can assist this process by identifying patterns in structured applicant data; however, predictive systems should provide understandable explanations rather than only a binary outcome. This paper presents KnowYourLoan, an explainable loan approval prediction and decision-support application based on CatBoost and SHapley Additive exPlanations (SHAP). The system estimates approval probability and applies a decision threshold of 0.45 to generate a predicted class. SHAP-based explanations communicate how applicant attributes contribute to individual predictions. The application uses a FastAPI backend, Supabase PostgreSQL, and web interfaces deployed through Cloudflare Pages. An administrative interface supports application record management, review status updates, notes, analytics, reports, and audit-related workflows. Project records describe a synthetic dataset of approximately 150,000 records and 35 features. Preliminary recorded evaluation metrics include ROC-AUC 0.8604, accuracy 77.66%, precision 76.66%, recall 85.80%, F1-score 80.97%, and Brier score 0.1506. These results must be reproduced against the final evaluation artifacts before publication. Since the data are synthetic, the reported results demonstrate experimental performance on generated data rather than validated performance on real loan applications.

Keywords—Loan approval prediction, CatBoost, explainable artificial intelligence, SHAP, machine learning, decision support, FastAPI.

1. Introduction

Loan approval is an important financial decision that requires the assessment of applicant characteristics and financial information. Conventional lending workflows may involve eligibility checks, document verification, and manual review. Machine learning provides a way to analyze structured data and estimate the likelihood of a specified outcome.

Tree-based methods can learn nonlinear relationships and feature interactions. However, a predictive score alone may not be sufficient for decision support: users and reviewers may need to understand which input attributes influenced a prediction. Explainable artificial intelligence (XAI) can help make model outputs more inspectable. SHAP assigns contribution values to input features, but explanations do not establish causality or guarantee fairness.

This paper presents KnowYourLoan, a web-based loan approval prediction system combining CatBoost classification with SHAP explanations. It receives applicant information, produces an estimated approval probability, and generates a predicted class using a configured threshold. It also provides an administrative interface for managing application records and reviewing their status.

The main contributions are CatBoost-based prediction; applicant-level SHAP explanations; a web interface for outcomes and probabilities; administrative workflows for records, status, notes, and reporting; and an explicit discussion of synthetic-data limitations. The system is experimental and is not an operational bank underwriting system.

2. Related Work

Machine learning has been widely applied to classification problems involving structured financial data. Logistic regression provides a comparatively simple baseline, while decision trees, random forests, and gradient-boosting methods can represent nonlinear decision boundaries. CatBoost is a gradient-boosting implementation designed to handle categorical features and reduce certain sources of prediction bias [1].

Ensemble models can be difficult to interpret directly. Global feature importance can indicate influential variables but does not necessarily explain an individual prediction. SHAP attributes model output to individual features using concepts from cooperative game theory [2]. TreeSHAP provides an efficient approach to computing attributions for tree-based models [3]. KnowYourLoan combines predictive modeling, local explanations, and administrative workflow; synthetic data limit conclusions about real-world credit decisions.

3. Dataset

Project records describe a synthetic dataset containing approximately 150,000 records and 35 features. Each record represents an applicant-like observation generated for model development and evaluation. The target is a binary loan approval outcome. Exact feature names, data types, value ranges, target-generation procedure, and class distribution should be confirmed from the final dataset and training pipeline.

Synthetic data can support software development without depending on confidential lender records, but may contain artificial relationships that do not reflect real-world lending patterns. Dataset size alone does not establish generalization or real-world validity. The positive class is treated as predicted approval in the deployed workflow, subject to verification against the training code.

TABLE I. DATASET SUMMARY

Attribute	Project description
Dataset type	Synthetic structured data
Approximate records	150,000
Approximate features	35
Prediction task	Binary classification
Model / explainer	CatBoost / SHAP TreeExplainer
Decision threshold	0.45

3.1 Input Features and Descriptions

Table II lists the 35 input features recorded in the project's current feature documentation. Before submission, feature names and definitions should be checked against the final model input schema and dataset columns.

TABLE II. INPUT FEATURES USED FOR LOAN APPROVAL PREDICTION

Feature	Description
Age	Applicant's age in years
Gender	Applicant's gender category
Marital Status	Applicant's marital status
Dependents	Number of financial dependents
Education	Applicant's education level
Employment Type	Applicant's employment category
Employment Years	Number of years employed
City Tier	Category of the applicant's city
Monthly Income	Applicant's monthly income
Co-applicant Income	Co-applicant's monthly income
Other Monthly Income	Additional monthly income
Existing Loan Count	Number of existing loans
Existing EMI	Monthly instalments of existing loans
Credit Card Outstanding	Outstanding credit card balance
Monthly Expenses	Applicant's monthly expenses
Total Outstanding Debt	Total outstanding debt
Credit Score	Applicant's credit score
Credit History Years	Length of credit history in years
Active Credit Accounts	Number of active credit accounts
Late Payments (12M)	Late payments during the past 12 months
Previous Defaults	Record of previous credit defaults
Credit Enquiries (6M)	Credit enquiries during the past six months
Credit Utilization	Proportion of available credit being used
Savings Balance	Applicant's savings balance
Property Value	Estimated value of property assets
Vehicle Value	Estimated value of vehicle assets
Investment Value	Value of financial investments
Loan Amount	Requested loan amount
Loan Tenure (Months)	Requested repayment period in months
Loan Purpose	Purpose of the requested loan
Annual Interest Rate	Annual interest rate associated with the loan
New Loan EMI	Calculated monthly instalment for the requested loan
Total Monthly Debt	Total monthly debt obligations including the new loan
DTI Ratio	Debt-to-income ratio
FOIR	Fixed Obligation to Income Ratio

4. Data Preprocessing

4.1 Input Validation and Transformation

Preprocessing prepares applicant records for consistent inference. Training and inference must use compatible transformations and the same expected feature names, types, and order. The transformation can be represented as $x' = T(x)$, where T denotes the application's preprocessing pipeline.

The workflow should validate required fields, convert numerical attributes, validate categorical values, handle missing or invalid values according to the training policy, exclude identifiers not intended as predictive inputs, and ensure that the inference schema matches the trained model. Actual missing-value treatment, categorical encoding, duplicate handling, and feature engineering should be documented from the final code.

5. Proposed System

KnowYourLoan comprises an applicant-facing interface, prediction API, trained CatBoost model, SHAP explanation module, application database, and administrative interface. The user submits applicant information; the backend validates it; CatBoost estimates the positive-class probability; the threshold produces a predicted class; SHAP calculates feature contributions; and the application displays the outcome and explanation. Application records can be managed through the administrative workflow.

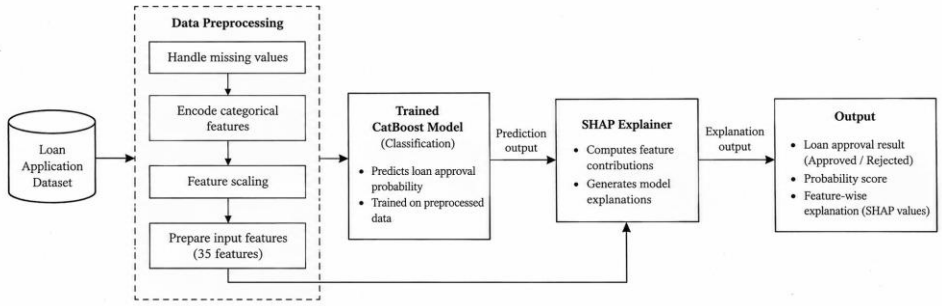


Fig. 1. Proposed system architecture of the KnowYourLoan application.

The system separates the interface, prediction logic, and persistent records. The frontend communicates with a backend API rather than executing the model directly in the browser.

6. Methodology

6.1 CatBoost Classification

CatBoost is a gradient-boosting algorithm based on decision trees. A simple ensemble representation is:

$$F(x) = \sum_{k=1}^{K} f_k(x)$$

where x is the input feature vector, f_k is the contribution of tree k, and K is the number of trees. For binary classification, the model estimates the probability $p(y = 1 \mid x)$.

The deployed project uses a recorded threshold of 0.45:

$$\hat{y} = 1 \text{ if } p(y = 1 \mid x) \geq 0.45; \text{ otherwise } \hat{y} = 0$$

This threshold is configurable and is not a universal lending standard.

6.2 SHAP Explainability

SHAP assigns contribution values to input features. For a model output on an appropriate explanation scale, $f(x) = \varphi_0 + \sum_{i=1}^{M} \varphi_i$, where φ_0 is the baseline, M is the number of input features, and φ_i is the contribution assigned to feature i. Interpretation depends on the output scale and explainer configuration. SHAP values describe model behavior; they do not prove that a feature causes approval or rejection.

6.3 Prediction Workflow

For each request, the application validates the expected schema, obtains the probability, applies the threshold, and computes explanation values for the same input. The response can be represented as $R(x) = [p(x), \hat{y}, \Phi(x)]$, where $\Phi(x)$ contains the explanation values.

7. Experimental Setup

The project uses Python-based machine learning components, CatBoost, SHAP, and a FastAPI backend. Supabase PostgreSQL is used for persistent application data, and web interfaces are deployed through Cloudflare Pages. Project resources include the customer interface at https://knowyourloan.pages.dev/, the administrative interface at https://knowyourloan.pages.dev/admin, API documentation at https://knowyourloan-api.onrender.com/docs, and the source repository at https://github.com/luqman19-idk/KnowYourLoan. These links are not independent evidence of model accuracy.

A reproducible setup should specify the exact dataset version, train/test split, stratification, random seed, CatBoost version, feature list, hyperparameters, threshold-selection procedure, and evaluation script. Unverified values are not assumed.

TABLE III. EXPERIMENTAL CONFIGURATION

Component	Recorded configuration
Classifier	CatBoost binary classifier
Explanation	SHAP TreeExplainer
Dataset	Synthetic; approximately 150,000 records
Feature count	Approximately 35
Threshold	0.45
API / database	FastAPI / Supabase PostgreSQL
Frontend hosting	Cloudflare Pages
Split / seed / hyperparameters	Verify from training code

8. Evaluation Metrics

Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives. Accuracy is (TP + TN)/(TP + TN + FP + FN). Precision is TP/(TP + FP). Recall is TP/(TP + FN). The F1-score is 2 × (Precision × Recall)/(Precision + Recall).

ROC-AUC evaluates the ability to rank positive observations above negative observations across thresholds. For binary outcomes, the Brier score is $\text{Brier} = (1/N) \sum_{i=1}^{N} (p_i - y_i)^2$. A lower Brier score indicates smaller squared probability errors on the evaluated dataset, but does not alone establish real-world calibration.

9. Exploratory Data Analysis

Exploratory analysis can inspect missing-value counts, duplicate records, feature distributions, categorical frequencies, target-class distribution, and relationships between numerical attributes. Target distribution should be examined because accuracy can be misleading when classes are imbalanced. Numerical features can be summarized using descriptive statistics and distribution plots; categorical features can be summarized using category counts. Correlation analysis provides a preliminary view of linear associations but does not establish causality or capture every nonlinear relationship.

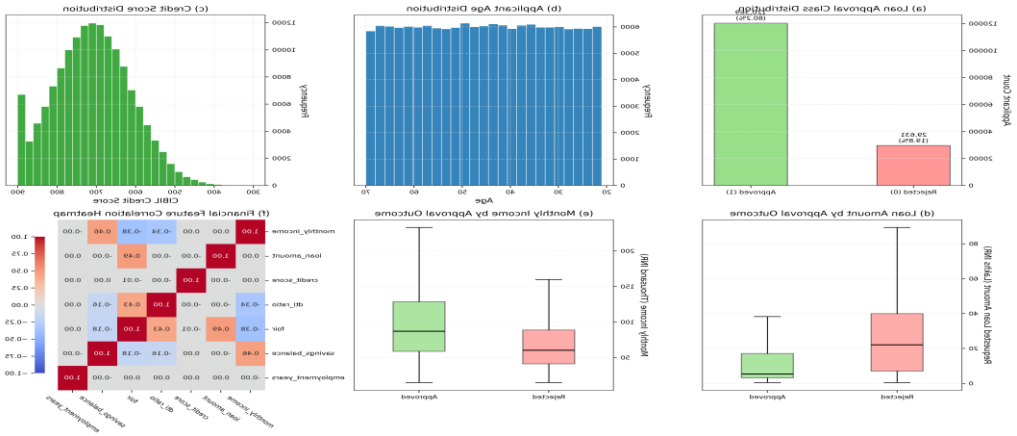


Fig. 2. Exploratory analysis of the synthetic loan dataset. Replace with verified plots from the actual training data.

10. Result Analysis

10.1 Classification Performance

Project records contain the following preliminary metrics for the CatBoost model. They must be reproduced against the final test set before publication.

TABLE IV. PRELIMINARY REPORTED CATBOOST PERFORMANCE

Metric	Reported value
Accuracy	77.66%
Precision	76.66%
Recall	85.80%
F1-score	80.97%
ROC-AUC	0.8604
Brier score	0.1506

The recorded recall indicates the proportion of positive-class observations identified in the evaluated dataset. Precision describes the proportion of predicted positives that were positive according to dataset labels. Accuracy summarizes overall correctness at the evaluated threshold. ROC-AUC summarizes ranking discrimination across thresholds, while Brier score summarizes squared probability error. Before publication, confirm that all metrics were generated from the same held-out test set and that threshold-dependent metrics use the documented 0.45 cutoff. Verify the positive-class definition, rounding, and Brier calculation.

10.2 Threshold Interpretation

The 0.45 threshold determines how probability becomes a class. A threshold below 0.50 can increase positive classifications relative to a 0.50 rule, depending on the probability distribution. Threshold selection should be justified using validation data and the intended application context.

10.3 Feature Importance

CatBoost feature importance was used to identify the input variables that contributed most to the trained model. The importance scores rank features according to the model's feature-importance measure. Higher scores indicate greater relative importance under this measure; however, they do not indicate the direction of influence or establish causal relationships.

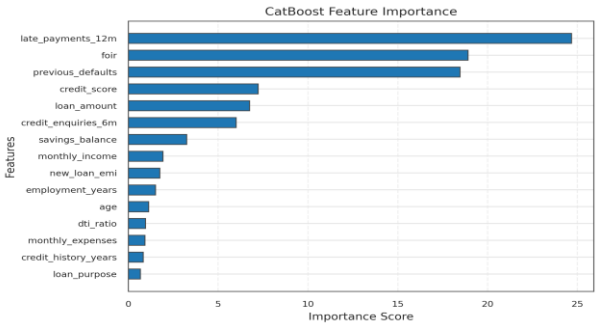


Fig. 3 - Feature Importance

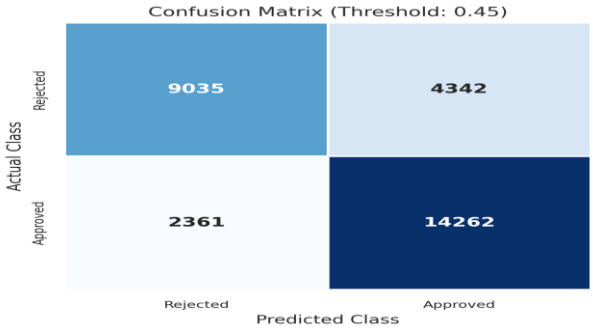


Fig. 4 - Confusion Matrix

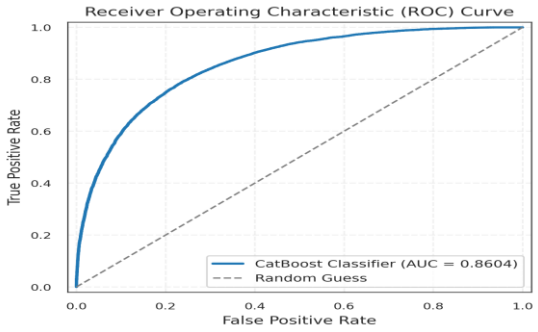


Fig. 5 - ROC Curve

11. Explainable AI Analysis

11.1 Global SHAP Analysis

Global SHAP analysis summarizes feature contributions across observations and can help researchers inspect which inputs influence model output. A summary plot can show the magnitude and direction of contributions for individual observations. A high contribution does not establish that a feature causes approval or rejection.

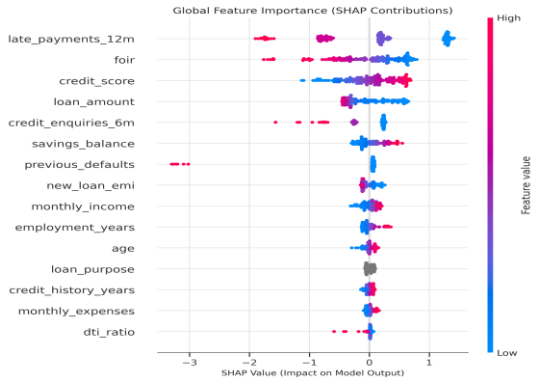


Fig. 6 – Global SHAP Summary Plot

11.2 Individual Prediction Explanation

Project records include an illustrative prediction with estimated approval probability of 85.8%. Under a 0.45 threshold, it would be assigned to the positive class. This is an individual example, not overall performance; its explanation should be reproduced from the saved model and corresponding input record before publication.

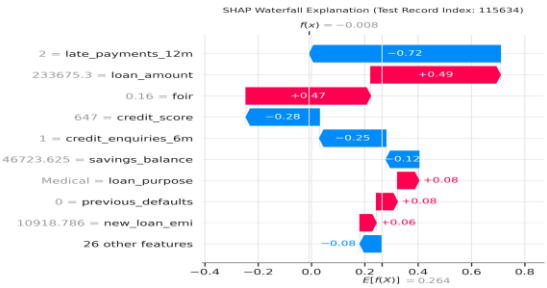


Fig. 7 - Verified individual SHAP waterfall or force-style explanation

11.3 Interpretation and Responsible Use

The explanation interface makes model behavior more inspectable, but does not independently validate correctness, remove bias, or provide a legally sufficient explanation for a real lender's decision. SHAP values should be described as model contributions, not causal reasons.

12. Sample Dataset Structure

The dataset is organized in a tabular format in which each row represents an applicant record and each column represents an input feature or the target variable. The input attributes describe demographic characteristics, employment, income, existing financial obligations, credit history, assets, and loan details. The target variable represents the loan approval outcome used for supervised classification. Table II illustrates the structure of the dataset.

TABLE V. ILLUSTRATIVE SAMPLE OF THE LOAN APPLICATION DATASET

Age	Credit Score	Monthly Income	Loan Amount	DTI Ratio	FOIR	Target
32	785	120,000	1,500,000	0.27	0.31	Approved
45	680	85,000	1,200,000	0.35	0.42	Rejected
29	740	95,000	800,000	0.24	0.29	Approved

Note: The values shown are illustrative and are provided only to demonstrate the dataset structure. The complete dataset contains 35 input features used by the loan approval prediction model, in addition to the target variable.

13. System Implementation

13.1 Customer Interface

The customer interface collects applicant information and sends it to the backend. It presents an estimated probability and threshold-based classification alongside influential input attributes. It should state that the output is an estimate from an experimental model, not a guarantee of a lender's decision.

13.2 Backend API

FastAPI receives requests, validates fields, invokes the trained pipeline, and returns prediction and explanation. Keeping inference in the backend supports a consistent implementation and avoids loading the trained model directly in the browser. Actual endpoint names, response schemas, authentication requirements, and API version should be documented from the deployed implementation.

13.3 Database and Administrative Workflow

The project uses Supabase PostgreSQL for persistent application data and includes a local SQLite/PostgreSQL database layer for development and deployment environments, according to project records. The administrative interface supports application record management, review queue, status updates, review notes, analytics, reports, and audit-related functionality. An administrator's review status should remain distinct from the model's predicted class: the first is an application workflow state, while the latter is generated by the classifier.

13.4 Deployment

The interfaces are deployed through Cloudflare Pages, the FastAPI backend on Render, and the database through Supabase PostgreSQL. A production-grade lending deployment would require further validation of access controls, privacy practices, monitoring, availability, model versioning, and data retention.

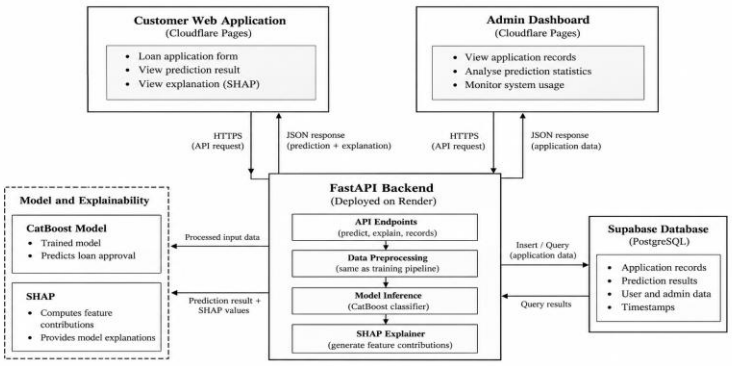


Fig. 8. Deployment architecture of the KnowYourLoan application.

14. Ethical Considerations

Loan-related predictions can influence access to financial opportunities. A model used in lending must be evaluated for reliability, fairness, privacy, and compliance. Synthetic data can encode artificial assumptions; performance on synthetic observations cannot establish performance on a real lending population.

Features may act as proxies for sensitive characteristics or reflect historical disparities. Fairness cannot be established merely by displaying SHAP values. Appropriate subgroup evaluation, legal review, and governance are needed before consequential use. Applicant information should be handled with appropriate security controls, data minimization, and access restrictions.

15. Limitations and Future Work

The main limitation is the use of synthetic data. A large synthetic dataset does not guarantee representative coverage of actual applicants or lending outcomes. The recorded metrics require reproduction from final evaluation artifacts. Further evidence is needed concerning calibration, subgroup performance, robustness to input changes, and threshold effects.

Future work should reproduce metrics from a documented held-out test set; publish the feature schema, preprocessing, class definition, split strategy, and hyperparameters; evaluate on independently sourced and appropriately governed lending data; compare CatBoost with suitable

baselines; examine calibration and error trade-offs; assess subgroup performance where lawful and appropriate; test explanation stability, correctness, and user understanding; and strengthen monitoring, model versioning, access control, auditability, and privacy safeguards. Independent validation and appropriate governance are necessary before considering the model for actual lending decisions.

16. Conclusion

This paper presented KnowYourLoan, a web-based loan approval prediction and decision-support application combining CatBoost classification with SHAP-based explanations. The framework generates a probability estimate, applies a recorded threshold of 0.45, and presents the resulting classification alongside feature-contribution information. It integrates a customer interface, FastAPI backend, PostgreSQL database service, and administrative workflow.

Project records report accuracy of 77.66%, precision of 76.66%, recall of 85.80%, F1-score of 80.97%, ROC-AUC of 0.8604, and Brier score of 0.1506. These figures must be reproduced from final evaluation artifacts before being treated as verified experimental results. The work demonstrates a practical way to combine tabular machine learning, explainability, and application workflows. However, synthetic data limit conclusions about performance, fairness, and reliability in real lending environments. The system remains an experimental prototype; independent validation, calibration assessment, fairness analysis, and governance are necessary before real-world use.

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